

# ACHINT PAL SINGH

## AI ENGINEER | RETRIEVAL, AGENTS & CLOUD SYSTEMS

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### PROFILE

AI engineer and Computer Science graduate building production AI applications, multimodal document pipelines, retrieval-augmented generation, and agent workflows. Hands-on ownership of Python/FastAPI services, model integration, evaluation, and GCP/Azure deployments. Focused on grounded answers, controlled execution, and reliable software delivery.

### CORE CAPABILITIES

**AI & retrieval:** RAG, FAISS, SentenceTransformers, CLIP, Tesseract OCR, LangChain, tool calling, model routing, session memory.

**Backend & data:** Python, FastAPI, Flask, SQL, PostgreSQL, MySQL, asynchronous REST APIs, Pandas, NumPy.

**ML & operations:** TensorFlow, scikit-learn, Docker, Terraform, GitHub Actions, pytest, GCP Cloud Run, IAM, Azure AI Search, Azure AI Foundry, monitoring.

### PROFESSIONAL EXPERIENCE

#### Predictive Tech Labs | AI Systems Engineer

Toronto, ON | July 2026 - Present

- Own backend and AI-platform engineering for WeKnowRights, building Python/FastAPI services, jurisdiction-aware RAG, semantic search, cited responses, and client-to-lawyer workflows.
- Engineer agent workflows with multi-provider model routing, tool calling, session memory, approval steps, and guarded execution.
- Build retrieval pipelines with FAISS, embeddings, metadata filtering, hybrid search, and grounding; evaluate quality, latency, cost, and reliability.
- Developed a two-tower recommendation architecture separating user and item representations for candidate retrieval.
- Deploy across GCP and Azure with Docker, Terraform, GitHub Actions, IAM, secrets management, structured logging, and repeatable releases.

#### Predictive Tech Labs | AI Engineer Intern

Toronto, ON | November 2025 - April 2026

- Built multimodal pipelines for PDF, DOCX, Excel, and images with Tesseract OCR, CLIP, SentenceTransformers, OpenAI embeddings, and FAISS.
- Created a generation and audit pipeline for six Ontario real-estate document types, combining curated templates, 54+ deterministic checks, and LLM semantic review.
- Reduced repetitive drafting by approximately four hours per closing workflow; achieved sub-100 ms latency in the deterministic audit layer.
- Implemented persistent conversation, case, and project memory with cross-user isolation and pytest end-to-end validation.
- Built and deployed a finance assistant using Azure AI Foundry, AI Search, Blob Storage, and App Service; validated pipelines through Azure DevOps.

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## SELECTED WORK & ADDITIONAL EXPERIENCE

### SELECTED PROJECTS

#### **WeKnowRights | Production legal AI platform**

- Integrated OCR, embeddings, FAISS retrieval, RAG Q&A, document generation, persistent memory, and lawyer-intake workflows behind asynchronous APIs.
- Evaluated model-routing trade-offs across cost, quality, latency, security, and reliability.

#### **Vector Search Benchmarking | Applied retrieval research**

- Benchmarked four open-source embedding models for retrieval quality and inference latency; documented practical model and vector-search trade-offs.
- Authored technical articles on vector search, RAG evaluation, tokenization, LLM economics, and AI-system design.

#### **Machine Learning & Mathematical Modeling**

- Built classification, regression, clustering, NLP, time-series, and deep-learning models using TensorFlow, scikit-learn, Pandas, and NumPy.
- Applied normalization, exploratory analysis, feature engineering, tuning, train/test validation, and comparative reporting.

#### **Personal Finance Management System**

- Built a full-stack application for transaction categorization, budgets, financial goals, and reporting; owned backend, database, interface, and deployment.

### ADDITIONAL EXPERIENCE

#### **Freelance | Full-Stack Developer & Data Analytics Consultant**

Toronto / Remote | August 2023 - Present

- Deliver small-business software and analytics from requirements and architecture through testing and deployment.
- Build web applications, RAG chatbots, and reporting workflows using Python, SQL, Power BI, and Tableau.

#### **Sudan Movers | Dispatcher & AI Solutions Contributor**

Canada | Part-time during studies

- Developed AI-assisted contract processing and coordination briefs from operational data.
- Used Python and Power BI for analytics, reporting, data quality, and fraud-detection experimentation.

### EDUCATION

#### **Algoma University | B.Sc. Computer Science**

September 2023 - April 2026

Coursework: machine learning, AI, deep learning, NLP, time-series analysis, databases, cloud computing, data structures and algorithms, Java, and agentic AI patterns.

### ADDITIONAL TOOLS

Java, JavaScript, Bash, PowerShell, Kubernetes, Cloud Build, Azure DevOps, BigQuery, Firestore, Azure Databricks, Power BI, and Tableau.